Actuator model of WFSim

I. Actuator

In Figure 1 we illustrate a two dimensional schematic representation of a turbine with yaw angle γ and wind direction at the turbine rotor defined by the angle ϕ.

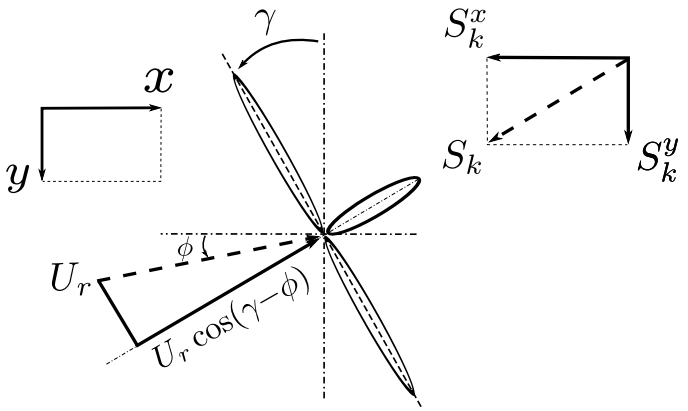


Fig. 1. Schematic representation of a turbine with yaw angle γ, wind direction angle ϕ and rotor velocity U_r.

According to momentum theory, the following force term can be defined:

$$S_k = C_T \frac{1}{2} \rho A_r U_{\text{effect}}^2, \tag{1}$$

with thrust coefficient C_T, rotor swept area A_r, air density $\rho = 1.2$ and rotor upwind velocity U_{effect}. In actuator disk theory, the thrust coefficient is defined as $C_T = 4a(1-a)$ with a the turbines axial induction, γ its yaw angle and ϕ the wind direction at the rotor. However, the authors in [**?**] show that for high axial inductions, this thrust coefficient is not accurate with respect to measurements and a Glauert correction is introduced. The following adaptation of the C_T definition is proposed in [**?**]:

$$C_T = \begin{cases} 4aF(1-a), & \text{if } 0 \leq a \leq 0.4 \\ \left(8/9 + (4F - 40/9)a + (50/9 - 4F)a^2\right) & \text{if } 0.4 < a < 1 \end{cases} \tag{2}$$

In this paper, we set $F = 2$. This does not only result in a relative increase of C_T for high axial inductions, but also for low axial inductions. Since U_{effect} is not well defined in a wind farm, it is interesting to write the force in terms of the rotor velocity. We define the following relations:

$$\beta = \frac{a}{1-a}, \qquad U_{\text{effect}} = \frac{U_r \cos(\gamma - \phi)}{1-a}, \qquad U_r = \sqrt{u_r^2 + v_r^2}, \tag{3}$$

with U_r the flow velocity vector at the rotor with direction defined by the angle ϕ. Note that u_r and v_r are velocity components at the rotor in x- and y-direction respectively. Substituting these relations in Eq.(1) yields the following expression for the force S_k:

$$S_k = \frac{1}{2} \rho C_T A_r \left[U_r \cos(\gamma - \phi)(\beta + 1)\right]^2 \tag{4}$$

Now we can define the force in the x- and y-direction as

$$S_k^x = -S_k \cos(\gamma), \qquad S_k^y = S_k \sin(\gamma) \tag{5}$$

II. MATLAB IMPLEMENTATION

```
function [Sm,Ueffect,Ur,a,Power,CT,phi,Fxmean,Fymean,Umean,Vmean] ...
    = Actuator(Wp,u,v,Phi,beta)


for kk=1:Wp.N

    a(kk)           = beta(kk)/(beta(kk)+1);

    vv              = diff(v(Wp.xline(kk,:),Wp.ylinev{kk}))/2+v(Wp.xline(kk,:)...
        ,Wp.ylinev{kk}(1:end-1));

    Umean(kk)       = mean(u(Wp.xline(kk,:),Wp.yline{kk}));
    Vmean(kk)       = mean(vv);

    % If we use alpha we do not get enough wake deflection
    alpha           = -180/pi*(atan(vv./u(Wp.xline(kk,:),Wp.yline{kk})));
    % Use this one instead ( will this give problems in derivatives ??? )
    phi(kk)         = -180/pi*mean(atan(vv./u(Wp.xline(kk,:),Wp.yline{kk})));

    U               = sqrt(u(Wp.xline(kk,:),Wp.yline{kk}).^2+vv.^2).*...
        cos((Phi(kk)-alpha)*pi/180);

    Ur(kk)          = mean(U);
    Ueffect(kk)     = Ur(kk)/(1-a(kk));

    %%
    F           = 2;

    if a(kk)> .4
        CT(kk)          = (8/9+(4*F-40/9)*a(kk)+(50/9-4*F)*a(kk)^2);
    else
        CT(kk)          =  4*a(kk)*F*(1-a(kk));
    end

    Fthrust          = 1/2*Rho*CT(kk)*(U*(beta(kk)+1)).^2;

    Fx              = Fthrust*cos(Phi(kk)*pi/180);
    Fy              = Fthrust*sin(Phi(kk)*pi/180);

    Fxmean(kk)      = mean(Fx'.*Wp.dyy2(1,Wp.yline{kk})');
    Fymean(kk)      = mean(Fy'.*Wp.dyy2(1,Wp.yline{kk})');


    Sm.x(Wp.xline(kk,:)-2,Wp.yline{kk}-1) = -Fx'.*Wp.dyy2(1,Wp.yline{kk})';
    scale                                 = (length(Wp.yline{kk}))/...
        (length(Wp.yline{kk})-1);
    Sm.y(Wp.xline(kk,:)-1,Wp.yline{kk}(2:end)-2) = ...
        Fy(2:end)'.*Wp.dyy2(1,Wp.yline{kk}(2:end))'*scale;

    %%
    Power(kk)       =  2*Rho*pi*(0.5*Wp.turbine.Drotor)^2*beta(kk)*...
        mean(U).^3;

end
```

III. Results

The following results are obtained for $u_b = 8$ and $v_b = 0$ so no side flow.

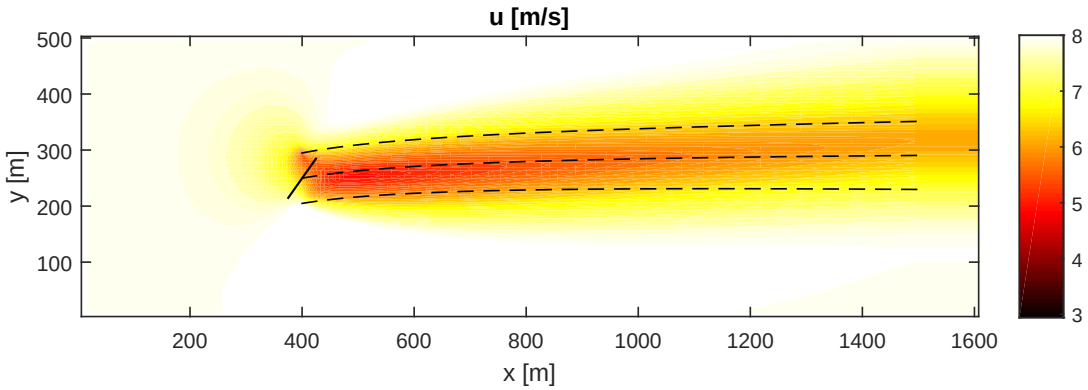


Fig. 2. Steady state u-velocity for a single turbine set up with $D_r = 90$ [m], $\gamma = 35$ [deg] and $\beta = \frac{1}{2}$ [m/s]. The black dotted lines are the theoretical wake centreline and wake diameter.

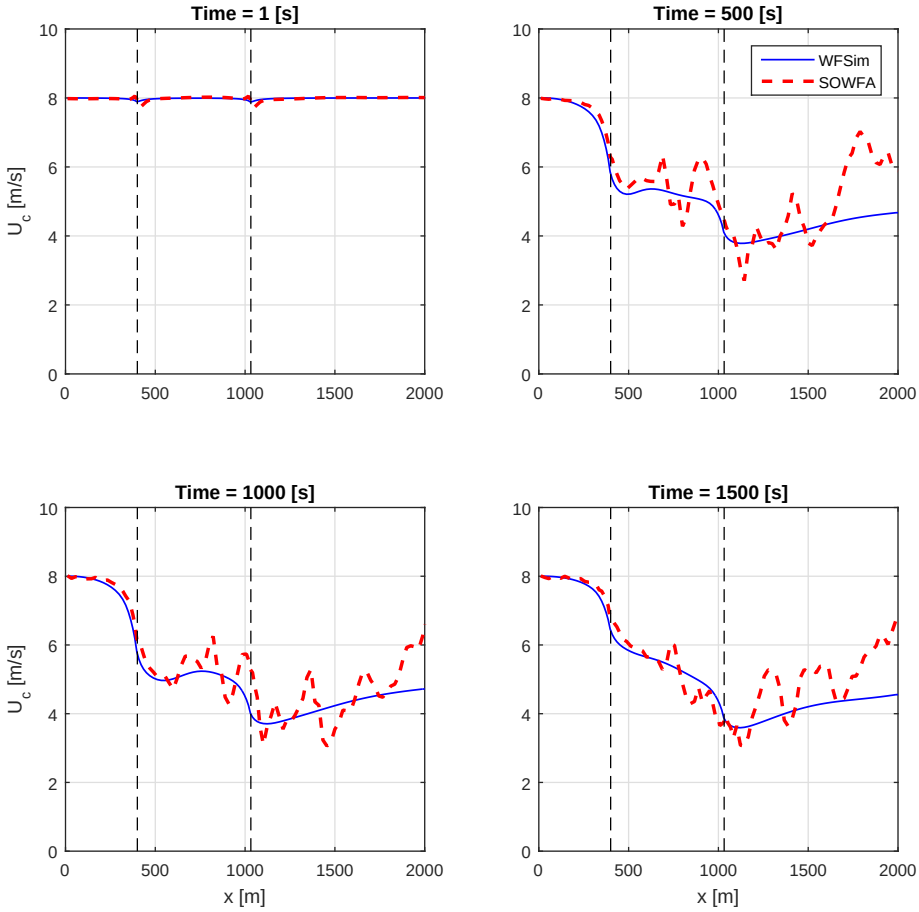


Fig. 3. Mean wake centreline U_c obtained with WFSim (blue) and SOWFA. The black dotted vertical lines give the turbine positions.

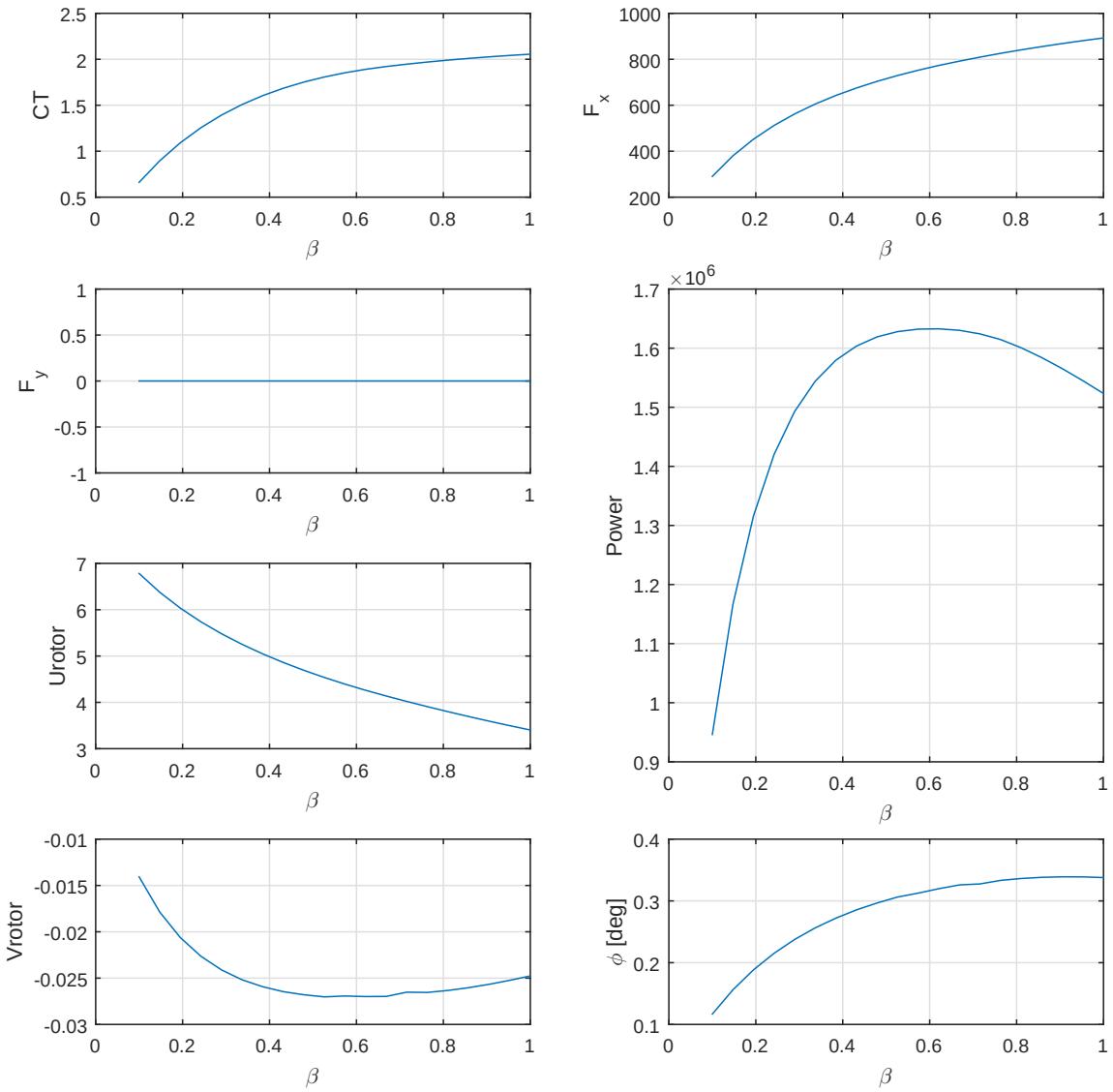


Fig. 4. Variables from WFSim for different β. Note that optimum can be achieved at $\beta = .5$ when setting $F = 2.3$ though the SOWFA comparison will become poor.

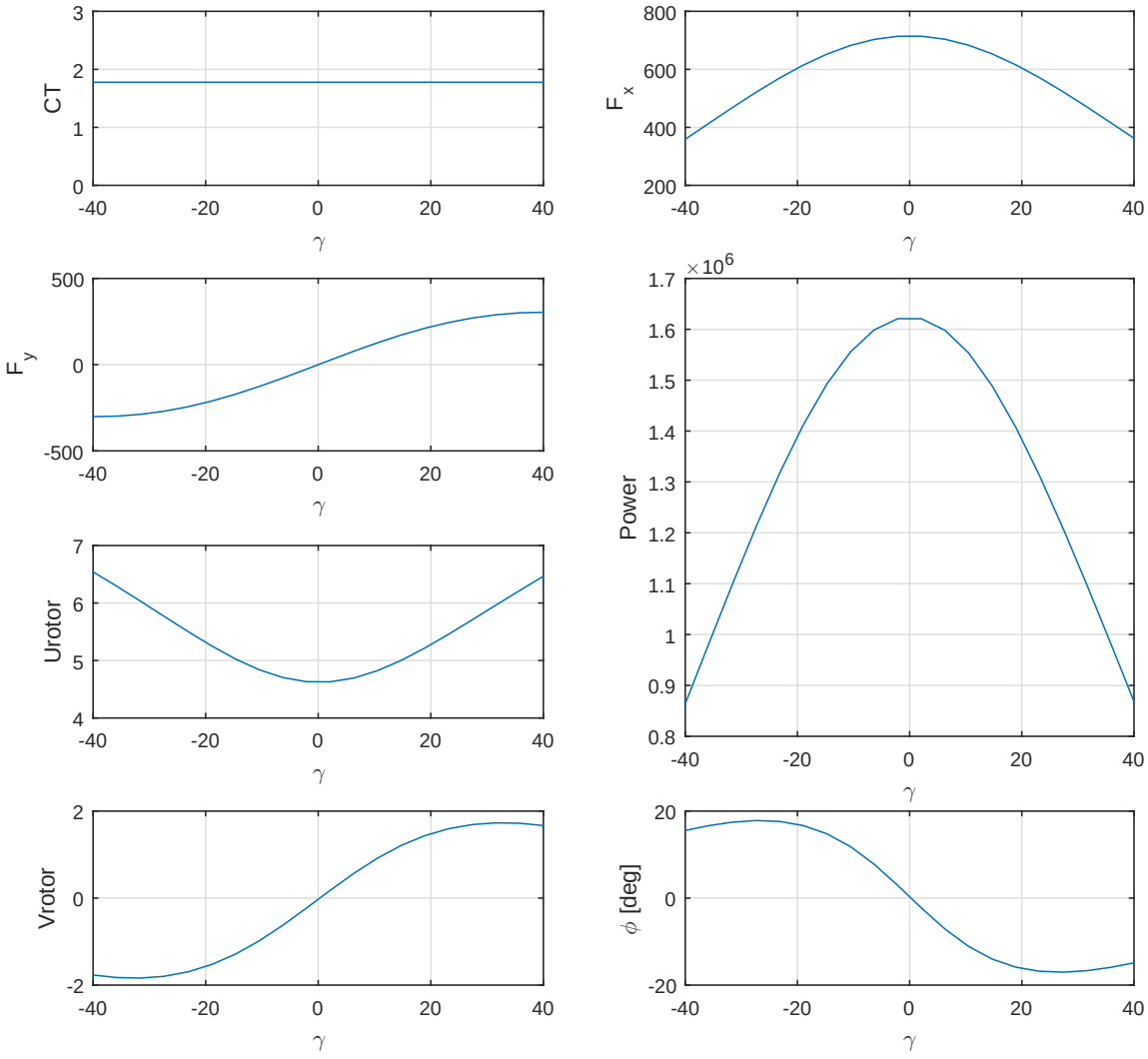


Fig. 5. Variables from WFSim for different γ.

IV. Derivatives

We had found

$$S_k = \frac{1}{2}\rho C_T A_r \left[U_r \cos(\gamma-\phi)(\beta+1)\right]^2, \tag{6}$$

with

$$C_T = \begin{cases} 4aF(1-a), & \text{if } 0 \le a \le 0.4 \\ \left(8/9 + (4F-40/9)a + (50/9-4F)a^2\right) & \text{if } 0.4 < a < 1 \end{cases} \tag{7}$$

The derivative to β is:

$$\frac{\partial S_k}{\partial \beta} = \rho A_r C_T(\beta)[U_r \cos(\gamma-\phi)]^2(1+\beta) + \frac{1}{2}\rho A_r [U_r \cos(\gamma-\phi)]^2(1+\beta)^2 \frac{\partial C_T(\beta)}{\partial \beta},$$

with

$$\frac{\partial C_T(\beta)}{\partial \beta} = \begin{cases} \dfrac{4F(1-\beta)}{(1+\beta)^3}, \\ \dfrac{(4F-40/9)(1+\beta)+2\beta(50/9-4F)}{(1+\beta)^3} \end{cases} \tag{8}$$

The derivative to γ is:

$$\frac{\partial S_k}{\partial \gamma} = -\rho C_T A_r \left[U_r(\beta+1)\right]^2 \cos(\gamma-\phi)\sin(\gamma-\phi) \tag{9}$$